Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Formalin is an aqueous solution of

(A) 40% formaldehyde

(B) 50% formaldehyde

(C) 60% formaldehyde

(D) 70% formaldehyde

Q.2

Which of the following has the most acidic hydrogen?

(A) 3-Hexanone

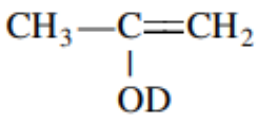
(B) 2, 4-Hexanedione

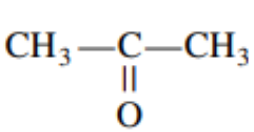
(C) 2, 5-Hexanedione

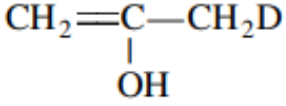
(D) 2, 3-Hexanedione

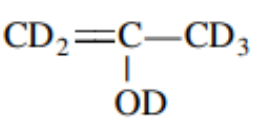
Q.3

The enol form of acetone, after treatment with D_2O, gives

(A) 

(B) 

(C) 

(D) 

Q.4

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When acetaldehyde is heated with Fehling's solution, it gives a precipitate of

(A) Cu

(B) CuO

(C) Cu_2O

(D) $Cu + Cu_2O + CuO$

Q.5

An oxime is formed when an aldehyde/ketone reacts with

(A) hydroxylamine

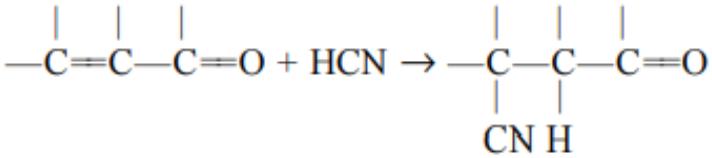
(B) hydrazine

(C) phenylhydrazine

(D) semicarbazide

Q.6

The reaction



involves

(A) electrophilic addition across double bond

(B) nucleophilic addition across double bond

(C) free radical addition across double bond

(D) no effect of the CO group on the $C == C$ bond

Q.7

Which of the following statements is correct?

(A) Benzaldehyde undergoes aldol condensation in an alkaline medium.

(B) Hydrazones of aldehydes and ketones are prepared in highly acidic medium.

(C) The reaction of methyl magnesium iodide with acetone followed by hydrolysis gives secondary butanol.

(D) The part of an aldehyde or a ketone immediately surrounding carbonyl carbon is flat.

Q.8

Acetophenone differs from aliphatic ketones in that it does not react with

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(A) HCN

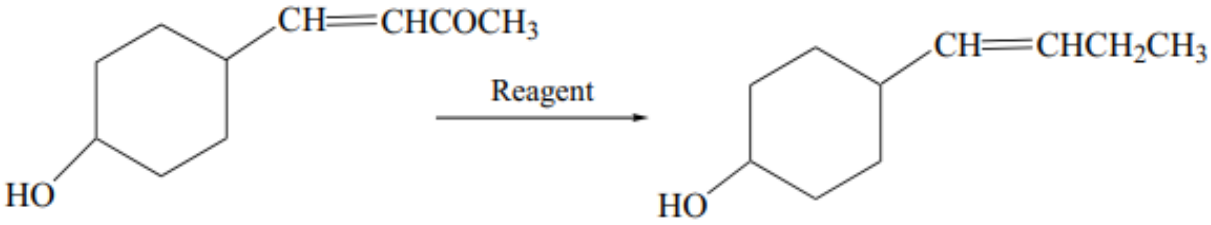
(B) $NaHSO_3$

(C) I_2 and $NaOH$

(D) NH_2OH

Q.9

In the given transformation, which of the following is the most appropriate reagent?



(A) NH_2NH_2, OH^-

(B) $Zn - Hg/HCl$

(C) Na, liq. NH_3

(D) $NaBH_4$

Q.10

Which one of the following reactions will not result in the formation of carbon - carbon bond?

(A) Reimer-Tieman reation

(B) Friedel Craft's acylation

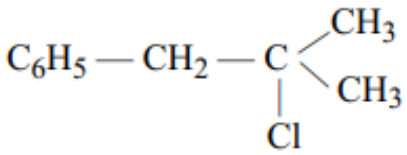
(C) Wurtz reaction

(D) Cannizzaro reaction

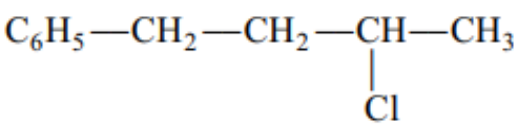
Q.11

A compound A with molecular formula $C_{10}H_{13}Cl$ gives a white precipitate on adding silver nitrate solution. A on reacting with alcoholic KOH gives compound B as the main product. B on ozonolysis gives Cannizaro reaction but not aldol condensation. D gives aldol condensation but not Cannizaro reaction. A is:

(A)



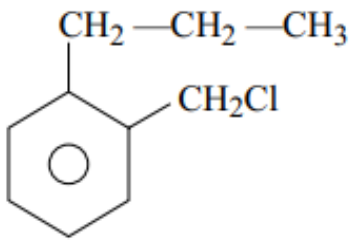
(B)



(C) $C_6H_5 - CH_2 - CH_2 - CH_2 - CH_2 - Cl$

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(D)



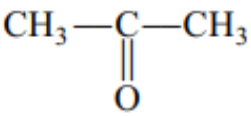
Q.12

In the reaction sequence

$2CH_3CHO \xrightarrow{OH^-} A \xrightarrow{\Delta} B$; the product B is:

(A) $CH_3 - CH_2 - CH_2 - CH_2 - OH$

(B) $CH_3 - CH = CH - CHO$

(C) 

(D) $CH_3 - CH_2 - CH_2 - CH_3$

Q.13

Trichloroacetaldehyde was subjected to Cannizaro's reaction by using $NaOH$. The mixture of the products contains sodium trichloroacetate and another compound. The other compound is

(A) chloroform

(B) $2, 2, 2$-trichloroethanol

(C) trichloromethanol

(D) $2, 2, 2$-trichloropropanol

Q.14

Acetone on reacting with CH_3MgBr followed by hydrolysis produces

(A) n-butyl alcohol

(B) sec-butyl alcohol

(C) isobutyl alcohol

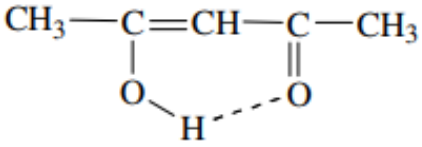
(D) tert-butyl alcohol

Q.15

Which of the following statements is correct?

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(A) The structure of the enol form of $CH_3COCH_2COCH_3$ with intramolecular hydrogen bonding is



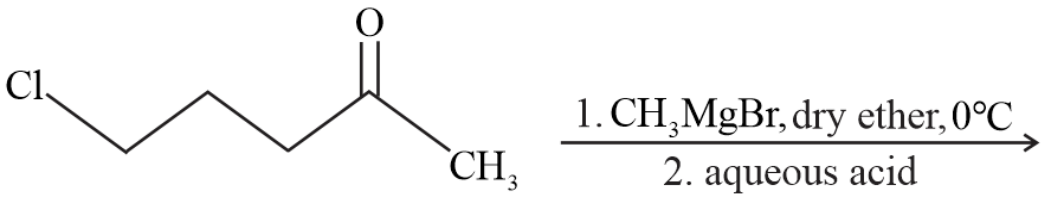
(B) Carbonyl carbon of an aldehyde or a ketone is sp hybridized.

(C) Fomaldehyde is a liquid at room temperature.

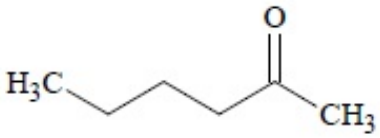
(D) Formaline is 30% aqueous solution of formaldehyde.

Q.16

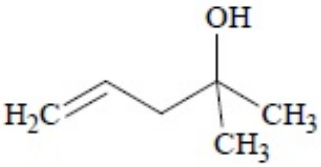
The major product in the following reaction is



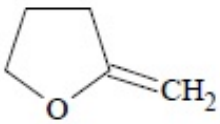
(A)



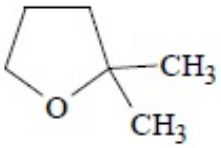
(B)



(C)



(D)



Q.17

An organic compound X is oxidized by using acidified $K_2Cr_2O_7$. The product obtained reacts with phenyl hydrazine, but does not answer silver mtextrror test. The possible structure of X is

(A) CH_3COCH_3

(B) $(CH_3)_2CHOH$

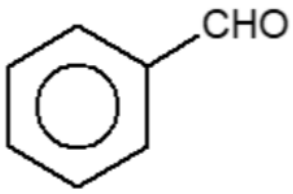
(C) CH_3CHO

(D) CH_3CH_2OH

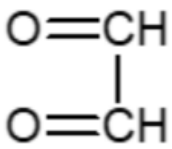
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Q.18

Which of the following will show cannizzaro reaction:

(A) 

(B) 

(C) 

(D) All of these

Q.19

Aldehyde with NH_2 - NH_2 forms

(A) Hydrazone

(B) Aniline

(C) Nitrobenzene

(D) None of these

Q.20

Which of the following can give iodoform test ?

(I) $CH_3 - \overset{\overset{O}{\|}}{C} - CH_2 - \overset{\overset{O}{\|}}{C} - CH_3$

(II) $C_6H_5 - CH_2 - \overset{\overset{O}{\|}}{C} - CH_3$

(III) $CH_3 - CHO$

(IV) $C_6H_5 - \overset{\overset{O}{\|}}{C} - CH_3$

(A) Only IV

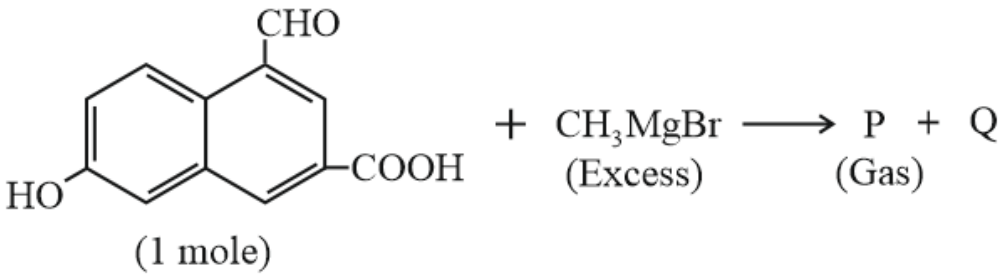
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(B) II and IV

(C) III and IV

(D) All of these

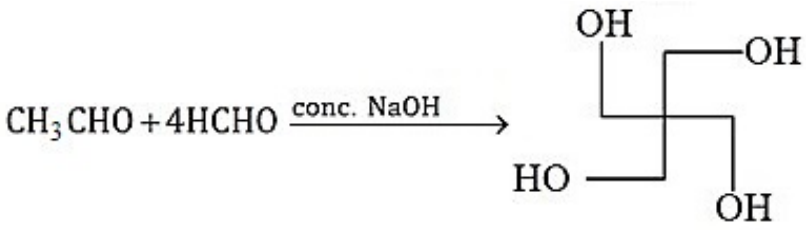
Q.21



How many litres of gas 'P' is formed in above reaction at NTP. (molar volume of gas at NTP is 22.4 L)

Q.22

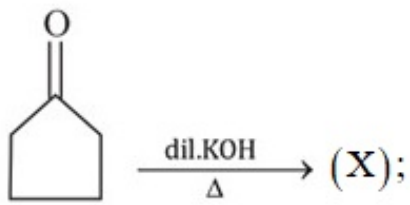
The number of aldol reaction (s) that occurs in the given transformation is



Q.23

The number of N - CH_2 - N bonds in urotropine is

Q.24



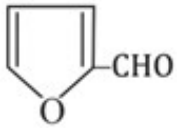
how many sp^2 carbon atoms are present in the final product?

Q.25

How many of these compounds can undergo cannizzaro reaction here

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CH_3CHO , CH_3COCH_3 , HCHO , Ph - CHO , Ph - CO - CHO ,



, $PhCOCH_3$

Answers & Solutions

Q.1 Answer:

40% formaldehyde

Solution:

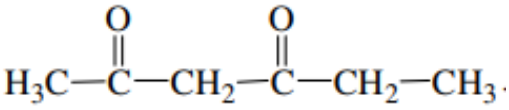
40% **formaldehyde is known as formaline.**

Q.2 Answer:

2, 4-Hexanedione

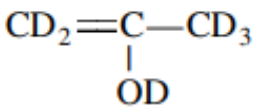
Solution:

2,4-Hexanedione is



Methylene hydrogens attached in between two $C = O$ **group will have maximum acidic nature.**

Q.3 Answer:



Solution:

The enol form of acetone, after treatment with D_2O **gives** $CD_2 = C(OD) - CD_3$**. All the** H **atoms of acetone are exchanged with** D **atoms during keto enol tautomerism in presence of** D_2O**.**

Q.4 Answer:

Cu_2O

Solution:

Red precipitates are of Cu_2O**.**

Q.5 Answer:

hydroxylamine

Solution:

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A aldehyde or a ketone containing $R-CO-$ group will produce an oxime on reaction with NH_2OH.

Q.6 Answer:

nucleophilic addition across double bond

Solution:

Addition across $C=O$ group occurs via nucleophilic mechanism.

Q.7 Answer:

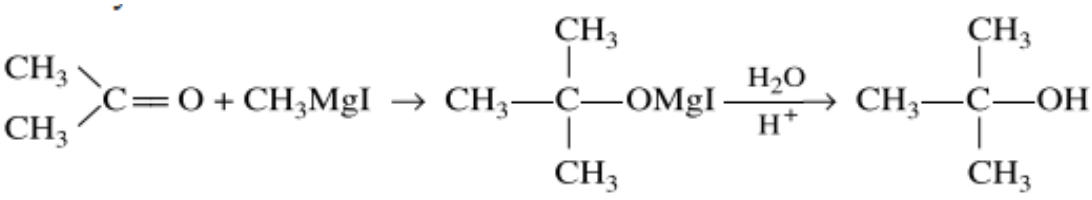
The part of an aldehyde or a ketone immediately surrounding carbonyl carbon is flat.

Solution:

(a) Benzaldehyde does not possess α-hydrogen atom. It does not undergo aldol condensation but undergoes Cannizzaro reaction giving benzoic acid and benzyl alcohol.

(b) Hydrazine gets protonated in highly acidic medium. Due to this, its nucleophilicity is decreased.

(c) The product is tertiary butanol.



(d) Carbonyl carbon is sp ^{2} hybridized.

So, the correct option is (d).

Q.8 Answer:

$NaHSO_3$

Solution:

$CH_3COC_6H_5$ does not react with $NaHSO_3$

Q.9 Answer:

NH_2NH_2, OH^-

Solution:

The Wolff-Kishner reduction (which uses hydrazine, NH_2NH_2, and a strong base OH^-) reduces only $-C=O$ group without affecting the $-C=C$ - group.

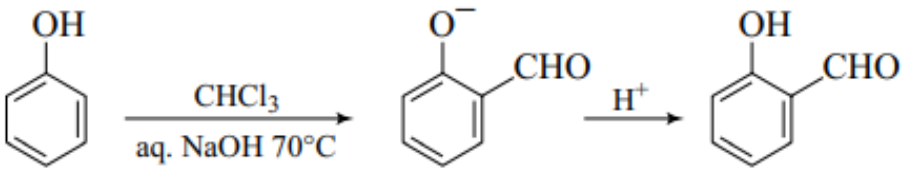
Q.10 Answer:

Cannizzaro reaction

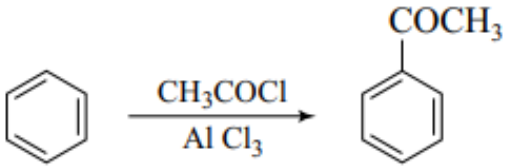
Solution:

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Reimer - Tiemann Reaction



Friedal - Craft's acylation



Wurtz reaction

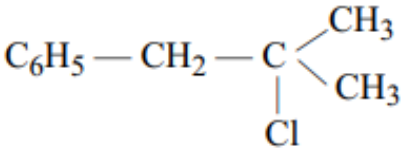
$$RBr + 2Na + BrR \rightarrow R - R + 2NaBr$$

Cannizzaro reaction

$$2HCHO \xrightarrow{50\%NaCH} CH_3OH + HCOO^-Na$$

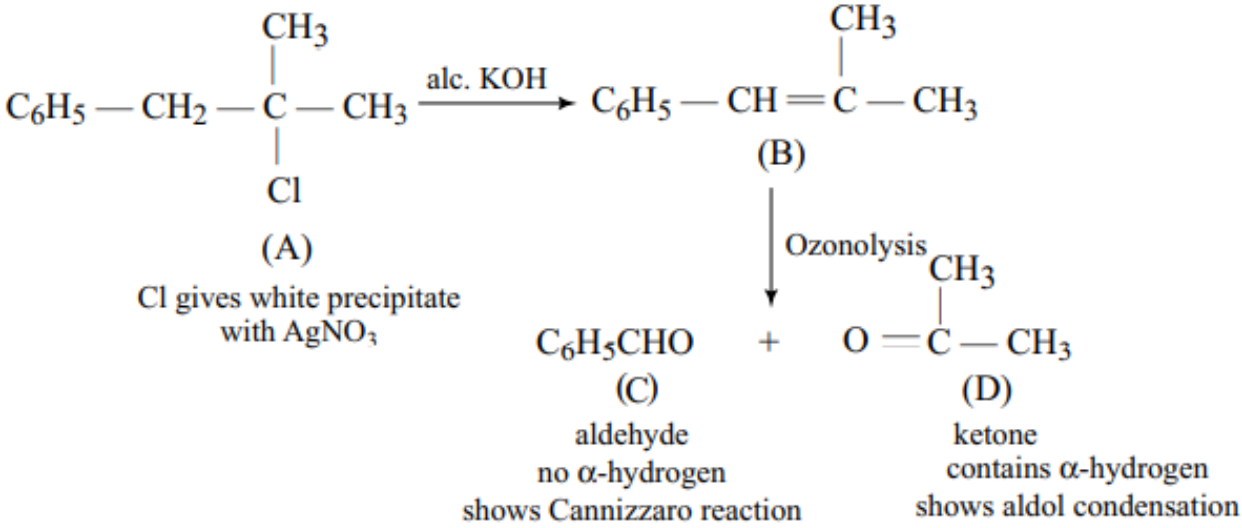
In Cannizzaro reaction, no C - C bond is formed

Q.11 Answer:



Solution:

The compound of choice (a) satisfies the given reactions.



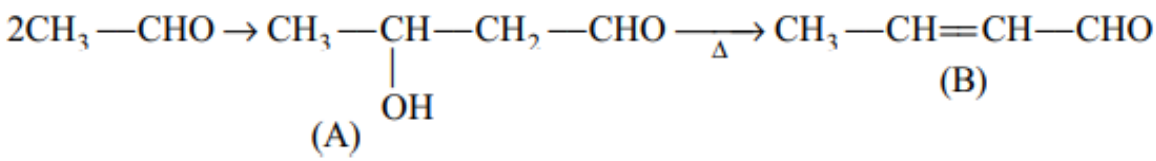
Q.12 Answer:

$$CH_3 - CH = CH - CHO$$

Solution:

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The reactions are:

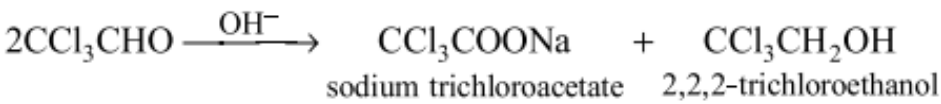


Q.13 Answer:

2, 2, 2-trichloroethanol

Solution:

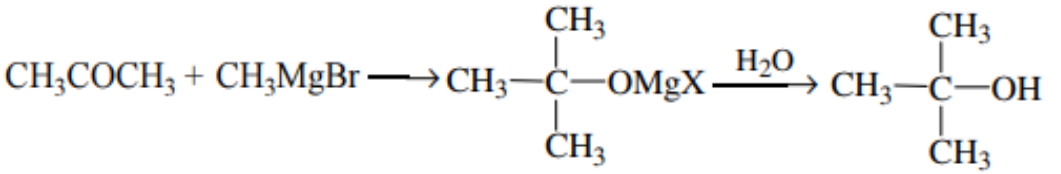
The reaction is



Q.14 Answer:

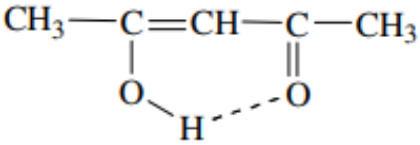
tert-butyl alcohol

Solution:



Q.15 Answer:

The structure of the enol form of $CH_3COCH_2COCH_3$ with intramolecular hydrogen bonding is



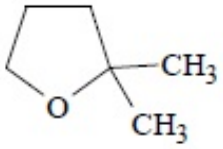
Solution:

(b) Carbon is sp ^{2} hybridized.

(c) Formaldehyde is a gas at room temperature.

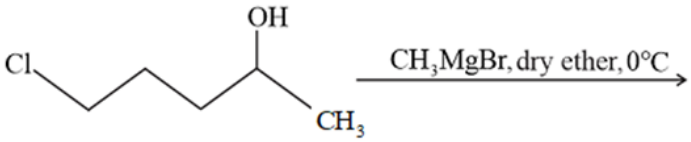
(d) Formaline is 40% aqueous solution of formaldehyde.So, the only correct option is (a)

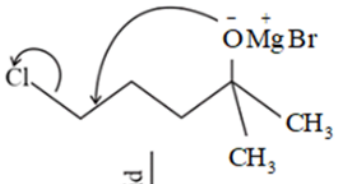
Q.16 Answer:



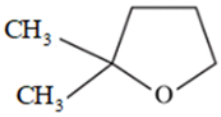
Solution:

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Q.17 Answer:

$CH_3{}_2CHOH$

Solution:

The oxidation product of X reacts with phenyl hydrazine, thus it contains $> C = 0$ group. The same product does not give silver mirror test. Thus, it is a ketone, because only aldehydes give silver mirror test.

Thus, the compound X must be 2° alcohol, as only secondary alcohols give ketones on oxidation and hence, X is $CH_3{}_2CHOH$.

Q.18 Answer:

All of these

Solution:

Carbonyl which does not have α - H can exhibit canizzaro reaction.

Q.19 Answer:

Hydrazone

Solution:

Aldehydes and ketones with NH_2 - NH_2 forms hydrazones. This is Nucleophilic additon followed by Elimination reaction

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$RCHO + H_2N - NH_2 \rightarrow RCH = N - NH_2 + H_2O$

Q.20 Answer:

All of these

Solution:

Anything that has

$CH_3 - \underset{\underset{O}{\|}}{C} -$

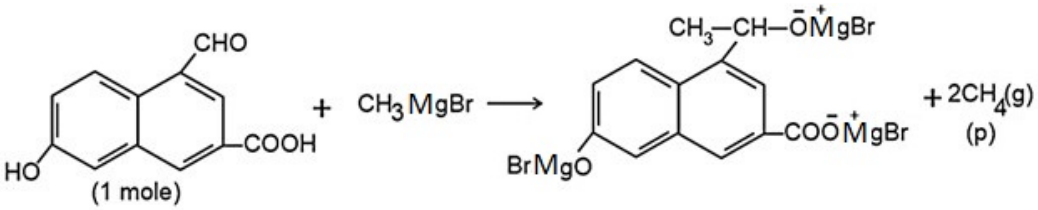
group gives positive iodoform

Q.21 Answer:

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Solution:

Grignard's reagent reacts with acidic-H to release methane



Volume of CH_4 at NTP $= 2 \times$ molar volume

$= 2 \times 22.4 = 44.8$ L

Q.22 Answer:

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Solution:

In this sequence of reaction three times aldol reaction occurs first than cross Cannizaro reaction takes place as follows

$CH_3 - CHO + 3HCHO \xrightarrow{OH^- \text{Aldol condensation 3times}} CH_2OH_3 - C - CHO$

$CH_2OH_3 - C - CHO \xrightarrow{OH^- \text{Cross cannizzaro reaction}} C - CH_2OH_4$

Q.23 Answer:

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Solution:

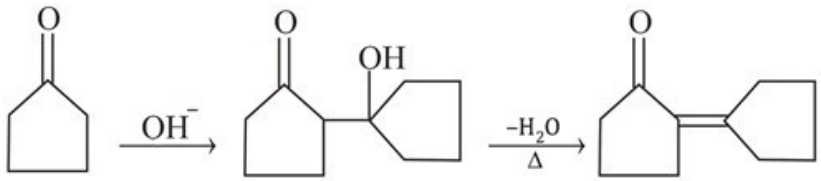
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Urotropine is hexamethylene tetramine $CH_{26}N_4$

Q.24 Answer:

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Solution:

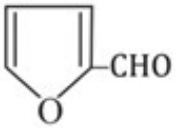


Q.25 Answer:

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Solution:

HCHO , Ph - CHO , Ph - COCHO **and**



can show cannizzaro reaction here.

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